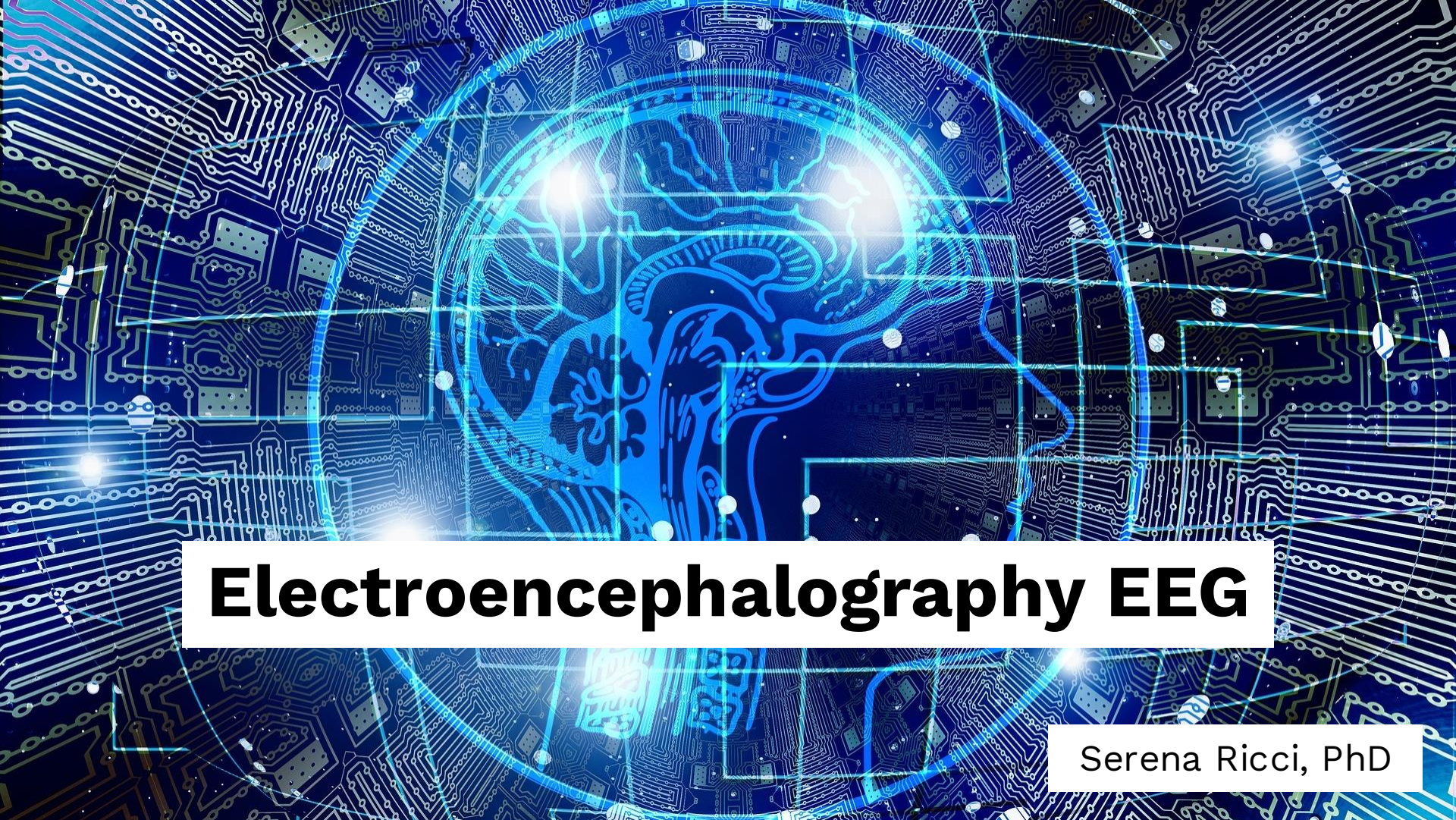
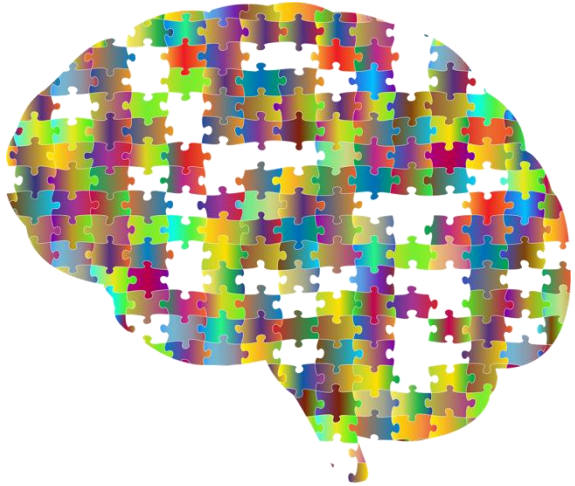


Electroencephalography EEG

Serena Ricci, PhD

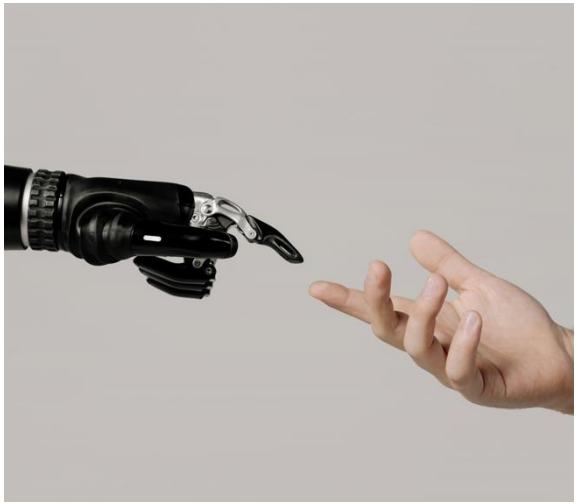
Topics

- The brain
  - Neurons
  - Synapse
  - Nervous system
- EEG
- EEG Analysis
- Preprocessing of EEG Data
- Hands on activity



Why EEG for robotics engineers?

- Development of Brain Computer Interfaces
  - Prosthetic control
  - Assistive Devices
- Human-robot interaction
  - Adaptive Robots
  - Collaborative Robots
- Neurofeedback
  - Cognitive load
  - Learning



The brain

The cells of the nervous system can be divided into:

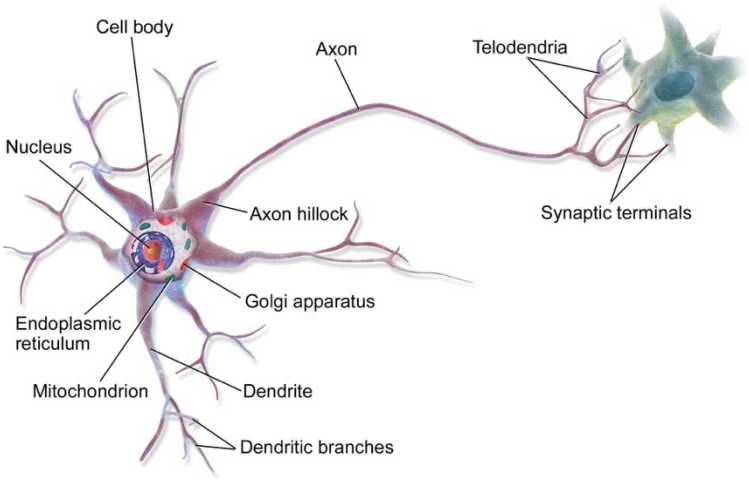
-Neurons -> specialized for electrical signaling over long distances

-Glia -> supporting cells not capable of electrical signaling

The human brain is estimated to contain 100 billion neurons
and several times as many supporting cells



Neurons



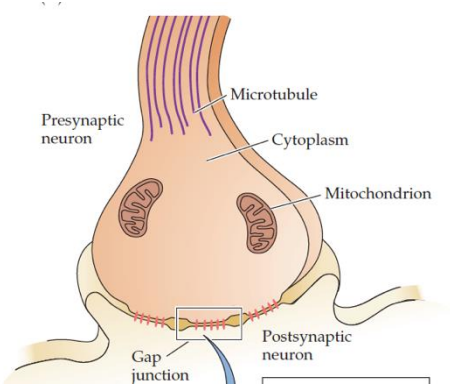
The basic cellular organization of neurons resembles that of other cells; however, **neurons** are specialized for intercellular communication.

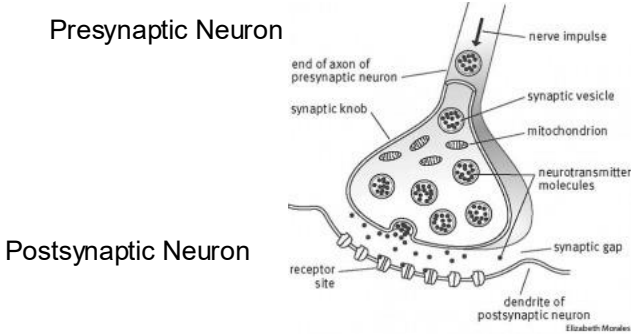
Neuron: electrically excitable cell that communicates with other cells via specialized connections (**synapses**)

Communication originates in the cell body, it transits in the axon up to the dendrites.

The number of inputs that a particular neuron receives depends on the complexity of its dendritic arbor

Synapse

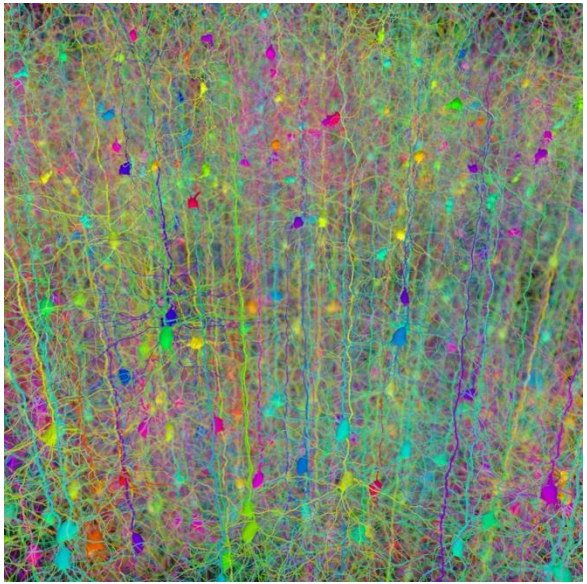




Electrical Synapse: Ionic current flows passively through the gap junction pores from one neuron to another.
Transmission can be bidirectional as current can flow in either direction across the gap junction and faster than chemical synapses

Chemical Synapse: junction between two neurons.
Neurotransmitter from the presynaptic cell activates ions channels on the post synaptic cell membrane so as ions flow in and out of the neuron causing a change in potential and then an electric field

Neural Circuits

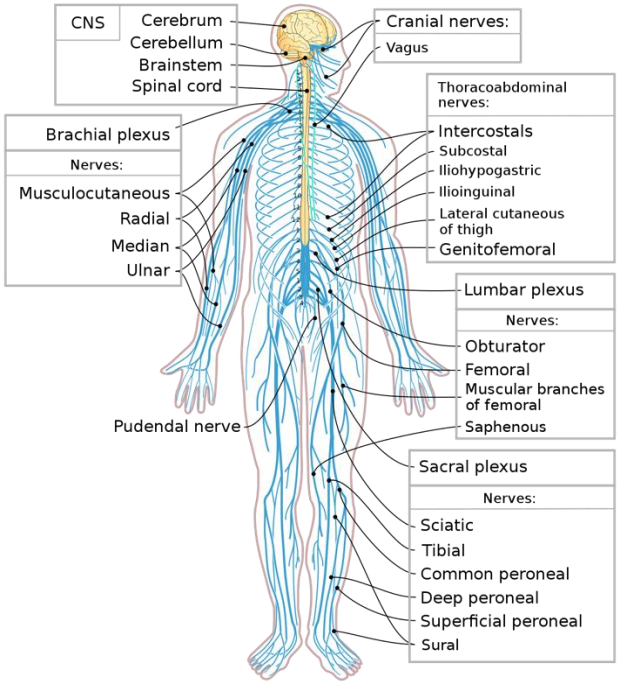


- The brain contains 100 billion neurons
- Neurons are organized into ensembles or neural circuits
- Neural circuits process specific kinds of information and provide the foundation of sensation, perception and behavior

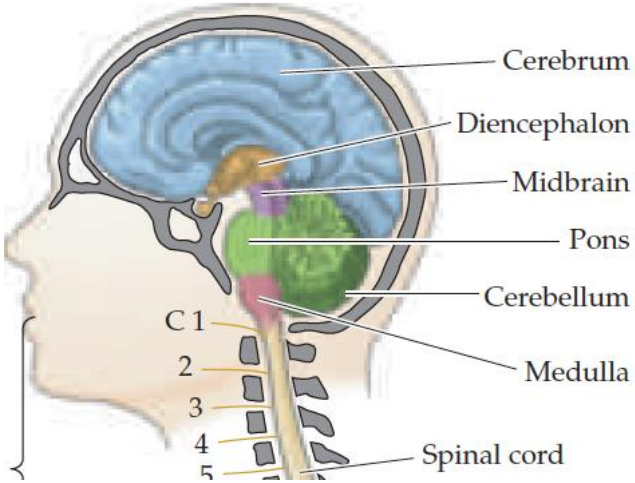
The nervous system

Central Nervous System: Brain and Spinal Cord

Peripheral Nervous System: Nerves



Central Nervous System

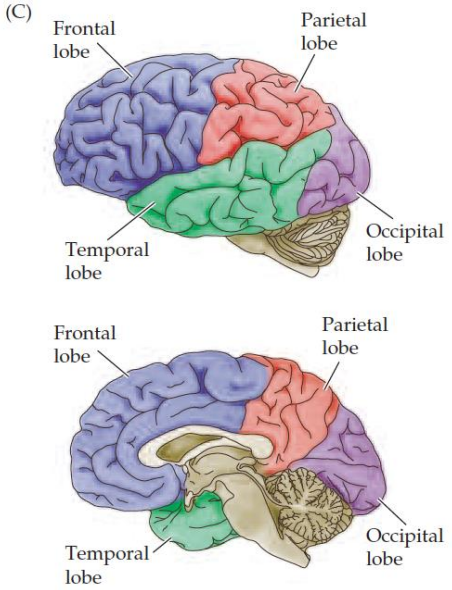


Cerebrum
Internal Structures
Spinal Cord

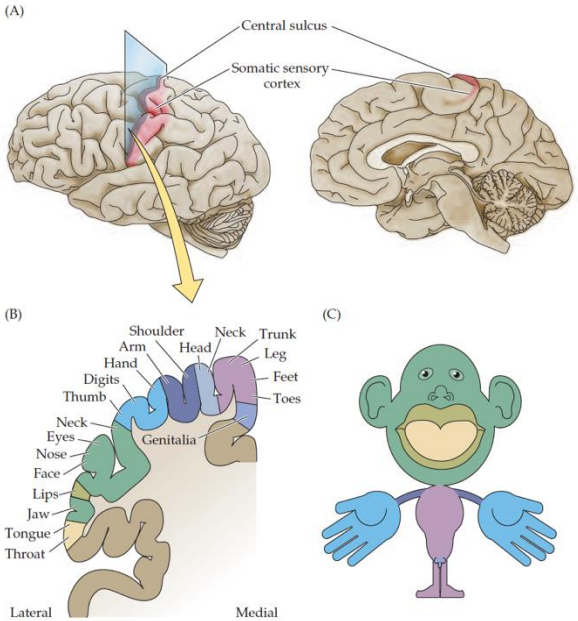
Cerebral hemispheres

Cerebral hemispheres can be divided into four lobes, according to the bones that overlie them:

- occipital
- temporal
- parietal
- frontal



Topographical organization



Map of the body surface that can be discerned within the various structures that constitute the somatic sensory system.
Adjacent areas on the body surface are mapped to adjacent regions.

This is important to understand how activity within the system provides a representation of relevant stimulus, the required response.

What is EEG?

EEG represents the electrical field of **thousand** of neurons simultaneously (neural population)

Such electrical activity can be recorded using **electrodes**



Invasive vs non-invasive





High density vs low density

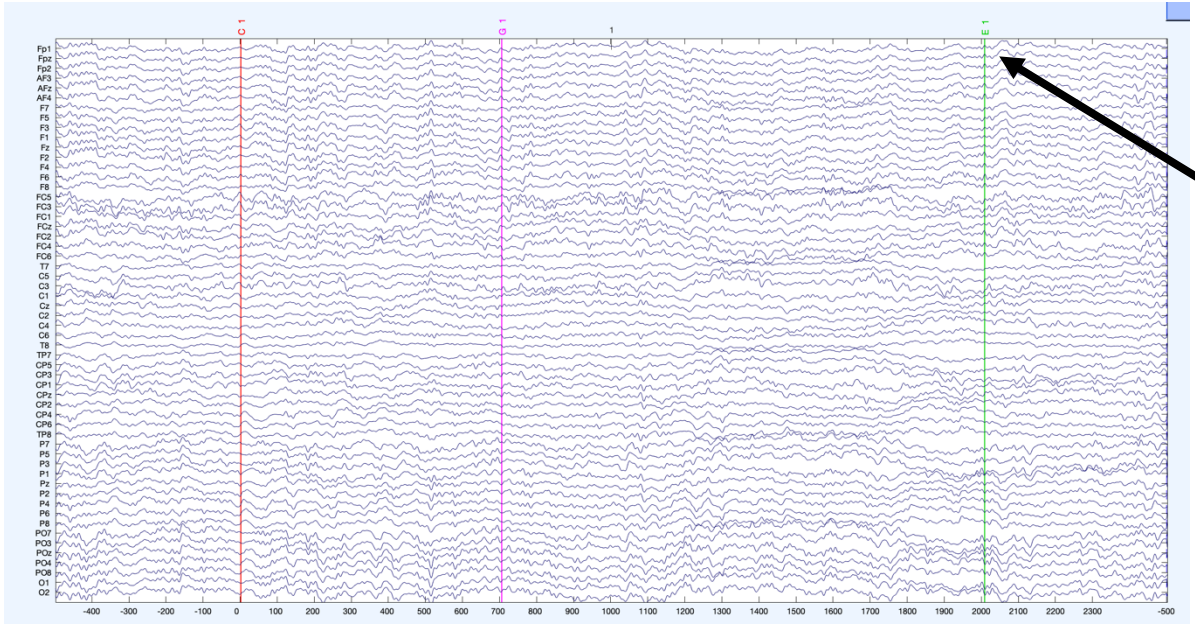
More than 64 channels -> High-density

Research: 32 to 256 channels

Clinics: 8 to 20 (max 32) channels



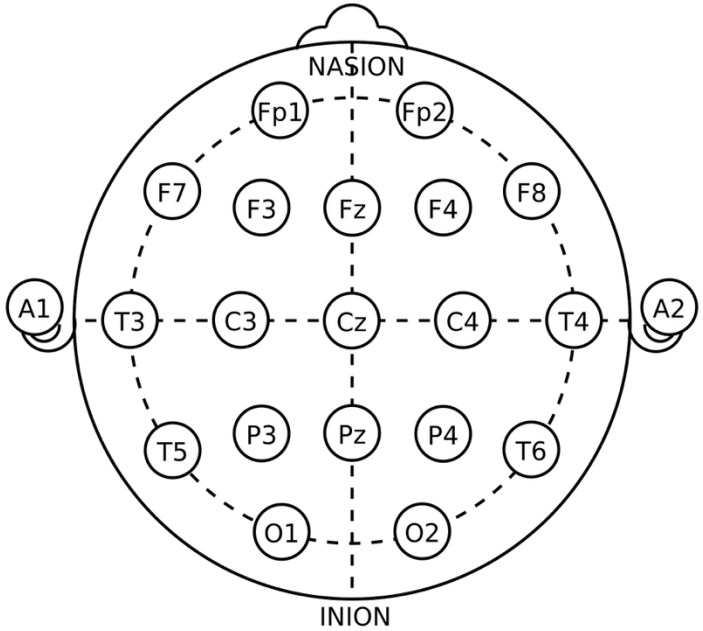
How does EEG look like?



Electrodes Placement–10/20system

Each electrode has a letter to identify the area of the brain it is reading from:

Pre-frontal (Fp), frontal (F), temporal (T), parietal (P), occipital (O) central (C).



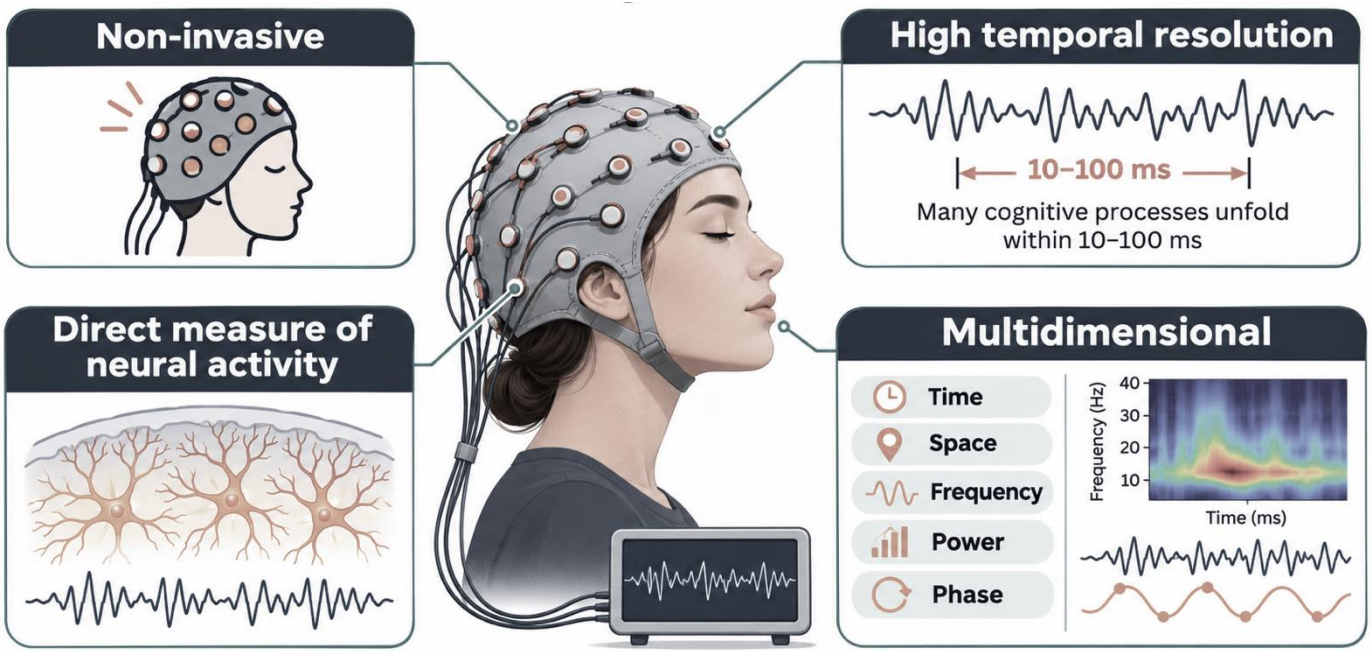
FpZ, Fz, Cz, Oz: electrode placed on the midline sagittal plane of the skull

Even numbers (2,4,6,8): right side
odd numbers (1,3,5,7): left side

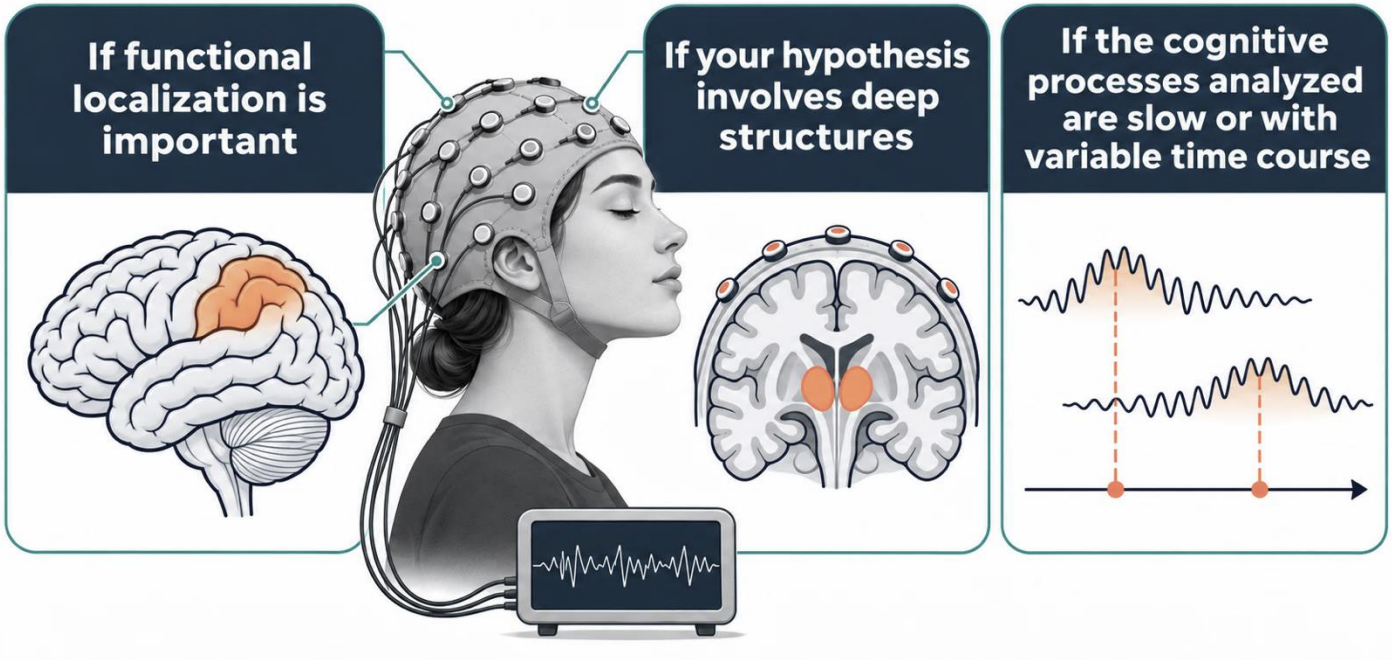
How is EEG measured?

- Activity recorded from one electrode is the change in the measured electrical potential between that electrode and a reference electrode-> **Relative measure**
- Electrical conductivity of air is almost zero -> a **physical electrical bridge** is required between scalp and electrode (gel)
- Electrodes can be wet (need electrolytic gel material as a conductor) or dry (single metal acting as a conductor)
- Unit of measurement -> volts
- μV value depends on subjects, recording, quality of the electrode and change when filters are applied -> Raw data are not usually informative

Why EEG?


Non-invasive
High temporal resolution
10–100 ms
Many cognitive processes unfold within 10–100 ms
Direct measure of neural activity
Multidimensional
Time
Space
Frequency
Power
Phase
Frequency (Hz)
40
30
20
10
Time (ms)

Why not EEG?

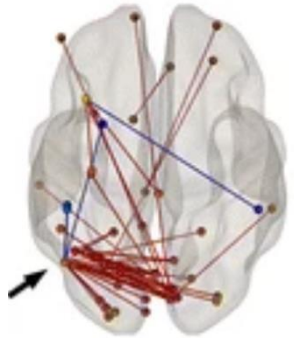

If functional localization is important
If your hypothesis involves deep structures
If the cognitive processes analyzed are slow or with variable time course

Type of Analysis

- Event-Related Potentials (ERP)
- Time Frequency analyses
- Non Linear Methods – Connectivity, Source

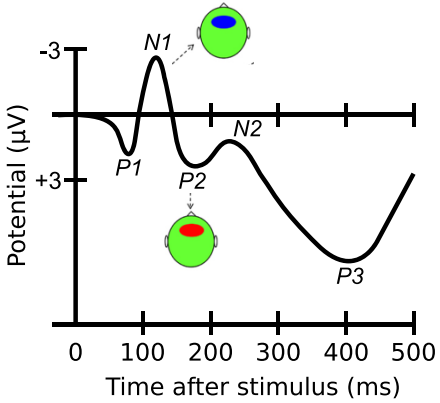
Non linear methods

Measures based on phase and power using a variety of linear and nonlinear methods to estimate connectivity between two signals



Event-Related Potentials

- ERP -> positive and negative polarity peaks at fixed intervals following the experiment
- High temporal precision and accuracy

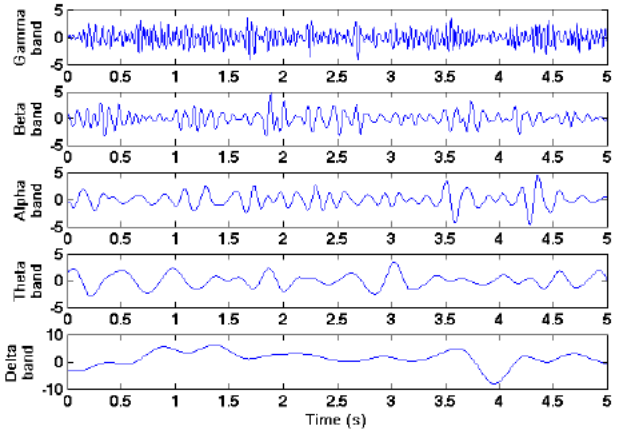


N1: between 80 and 120 milliseconds after the onset of and auditory stimulus.

P2: ~ 200 after the onset of a visual stimulus.

Time Frequency

- Oscillations -> rhythmic fluctuations in the excitability of neuron populations due to neurobiological events
- Analyzed with time-frequency-based approaches



Gamma 30 – 100 Hz –> higher cognitive functions, i.e. memory/learning

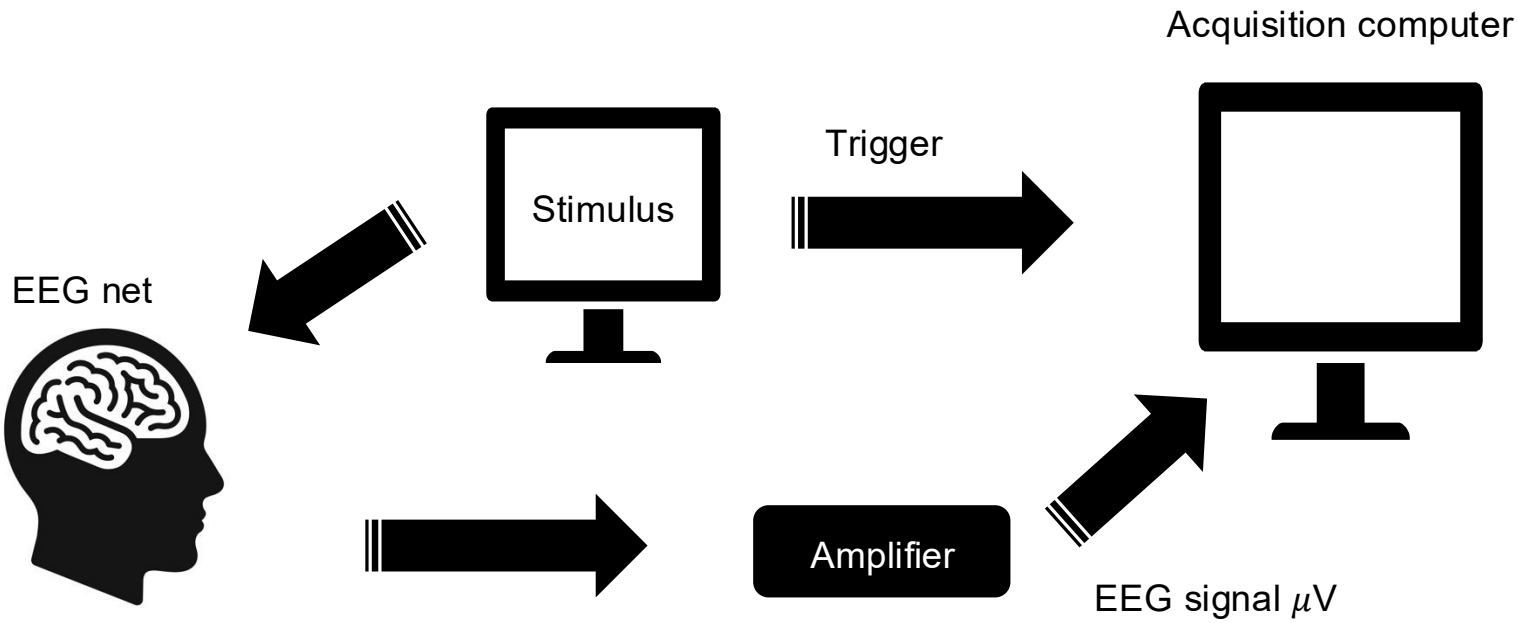
Beta 14/15 – 25/30 Hz -> movements

Alpha 7/8 – 13 Hz -> eyes closed

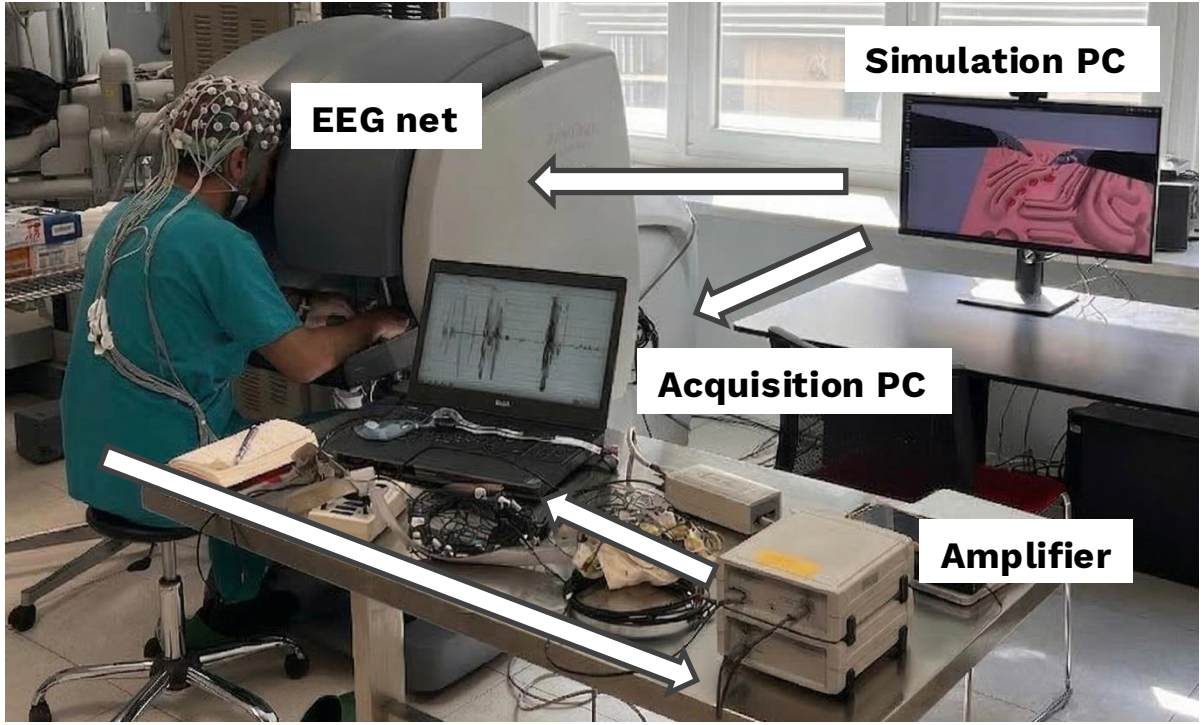
Theta 4– 7/8 Hz -> attention

Delta: 0.5 – 4 Hz -> sleep

Acquisition System



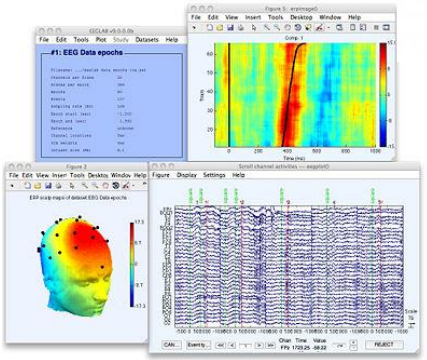
Acquisition System



Toolboxes for EEG Data analysis

- EEGLAB
- Brainstorm
- Fieldtrip
- pyEEG
- Brain Vision Analyzer
- ...





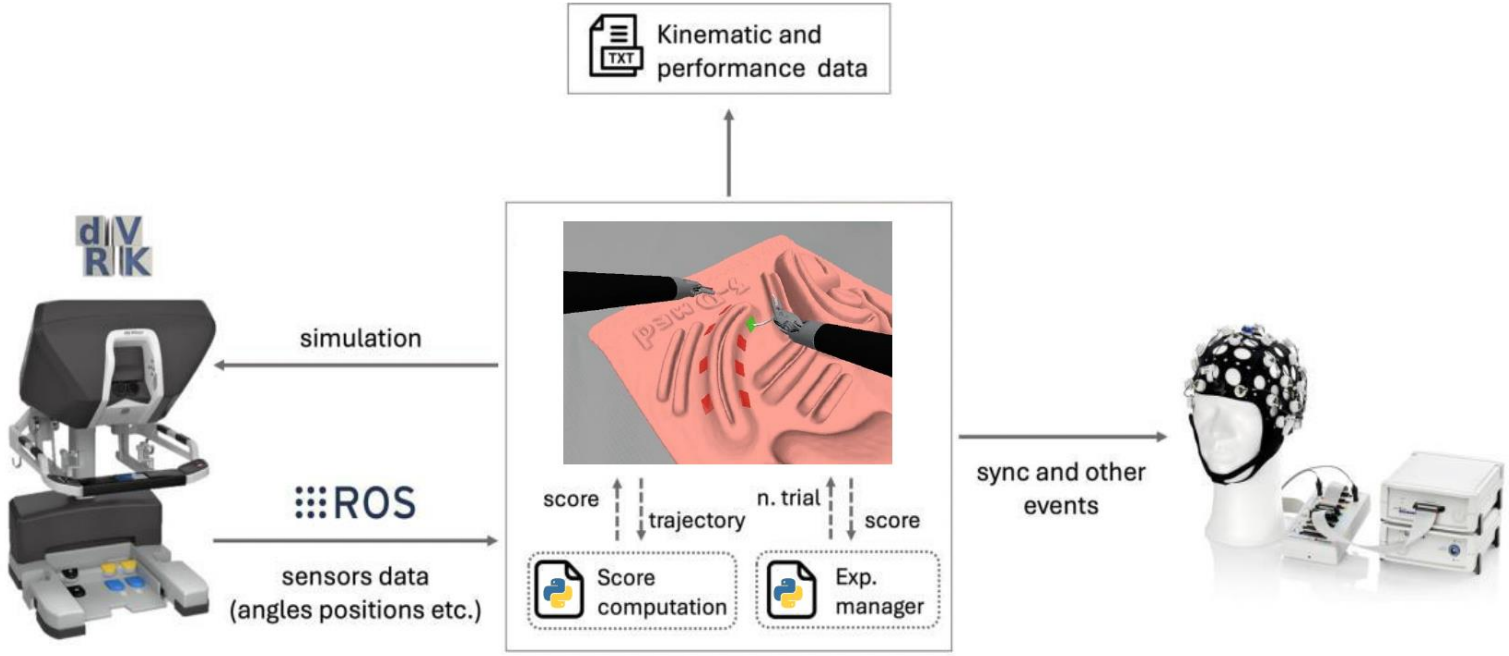


EEG preprocessing

Hands-on activity



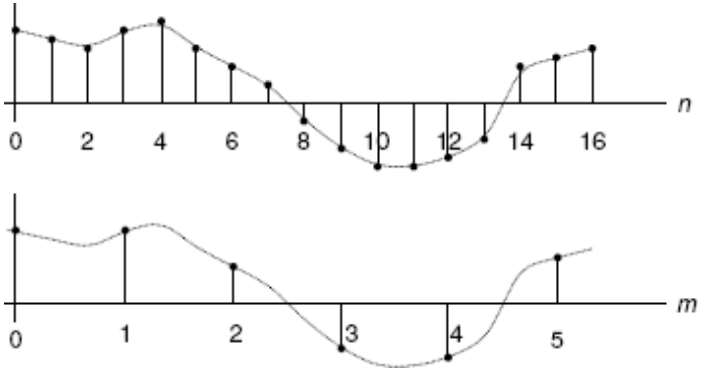
Experimental Setup



Preprocessing

0. Channels location (if necessary)
1. Downsampling (if necessary)
2. Filtering
3. Epoching
4. Visual inspection for channels and epochs
5. Independent Component Analyses
6. Channels Interpolation
7. Average Reference

Downsampling



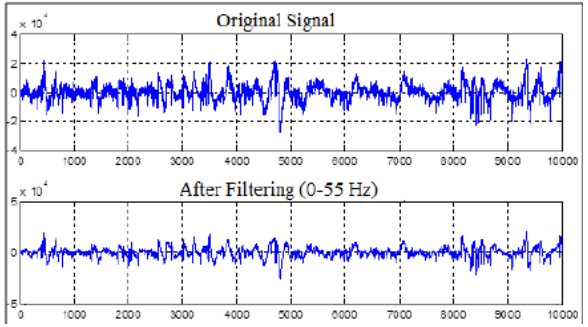
Theory: you need to sample at least twice the highest frequency of interest (Nyquist)

Practice: you should sample more than twice the highest frequency of interest to increase signal to noise ratio. Acceptable sampling frequency are between 250 and 500 Hz.

Filtering

In order to visualize the data it is good practice to filter them

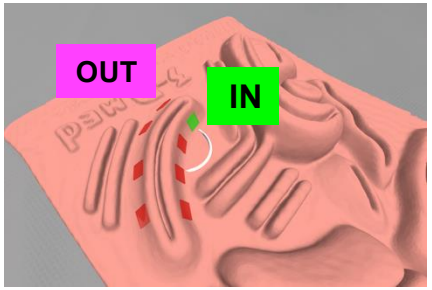
- High Pass 1/2Hz
- Low Pass 80 Hz
- Notch Filter 50 Hz

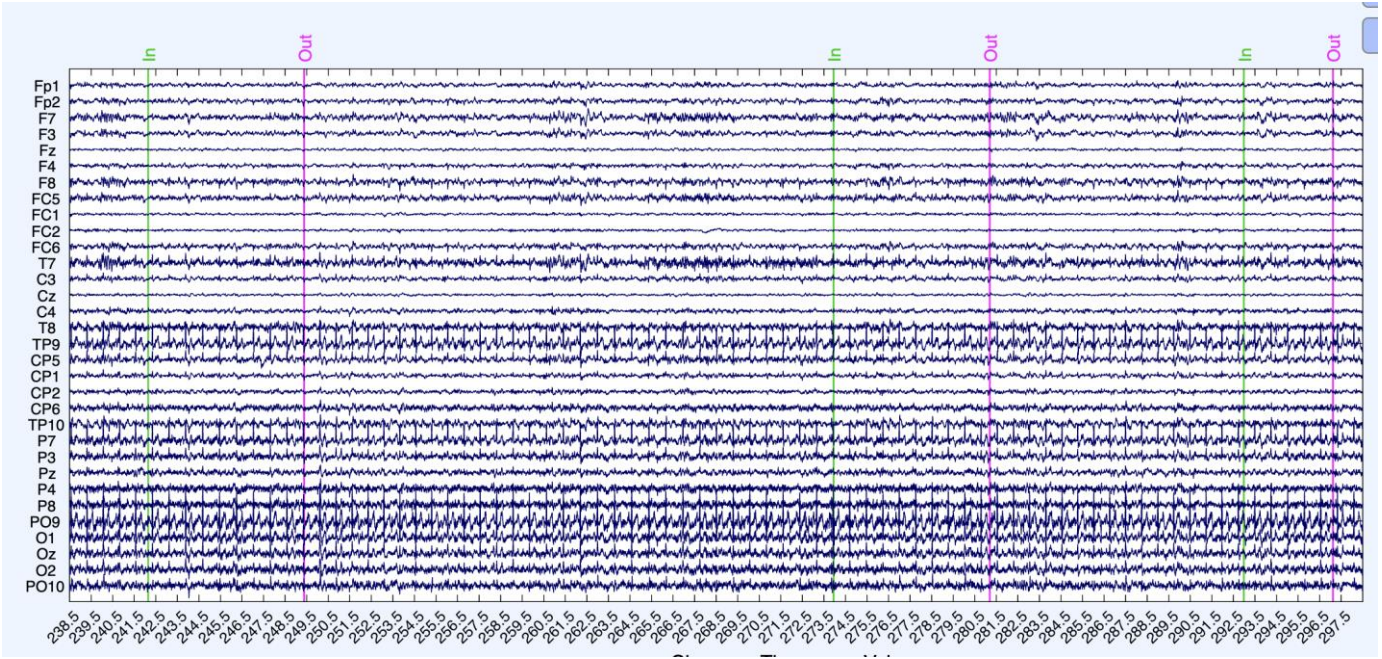


Finite Impulse Response (FIR)

Filter Order: $\text{ceil}(\frac{3.3}{\frac{FilterEdge*0.25}{Sampling\ Freq}})$

Epoching


OUT
IN

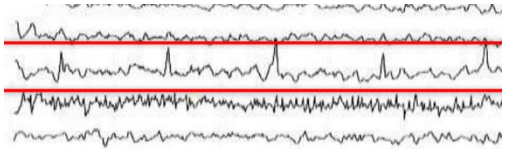

In
Out
In
Out
In
Out
Fp1
Fp2
F7
F3
Fz
F4
F8
FC5
FC1
FC2
FC6
T7
C3
Cz
C4
T8
TP9
CP5
CP1
CP2
CP6
TP10
P7
P3
Pz
P4
P8
PO9
O1
Oz
O2
PO10

Visual Inspection

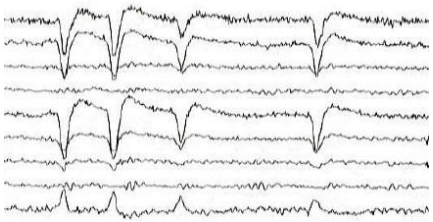
- Although EEG is designed to record cerebral activity, it also records electrical activities arising from sites other than the brain.
- Artifacts can be divided in physiologic artifacts (generated from the subject) and extra physiologic artifacts arise from outside the body
- Recognize artifact is important process to reduce the chance of misinterpretation of the EEG

Physiologic Artifacts

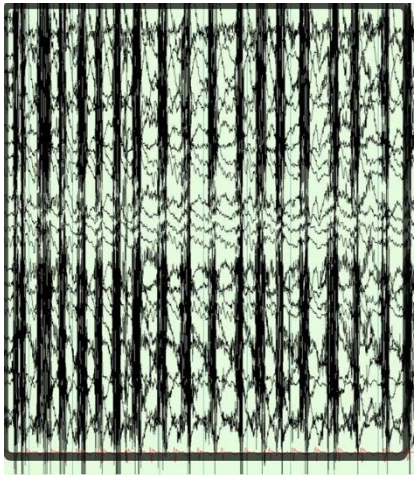
Cardiac Artifact



Eyes Blink

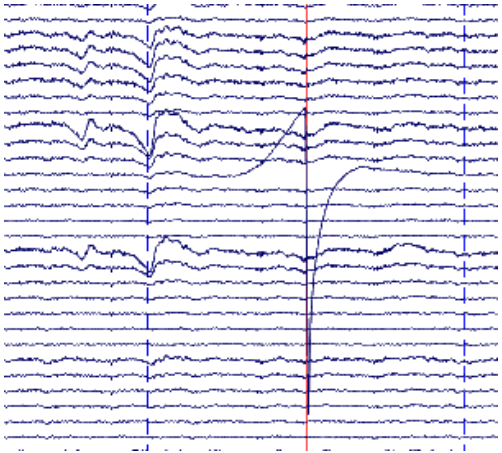


Muscle Artifact

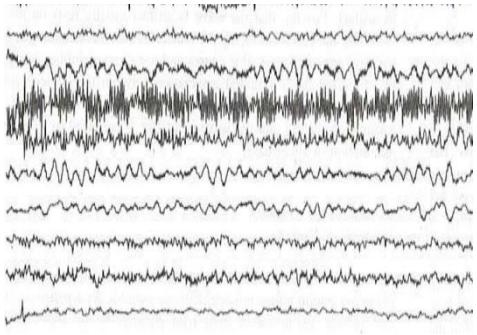


Extraphysiologic Artifacts

Electrode Artifact



50 – 60 Hz Noise



What to do with artifacts?

Physiologic -> leave it there

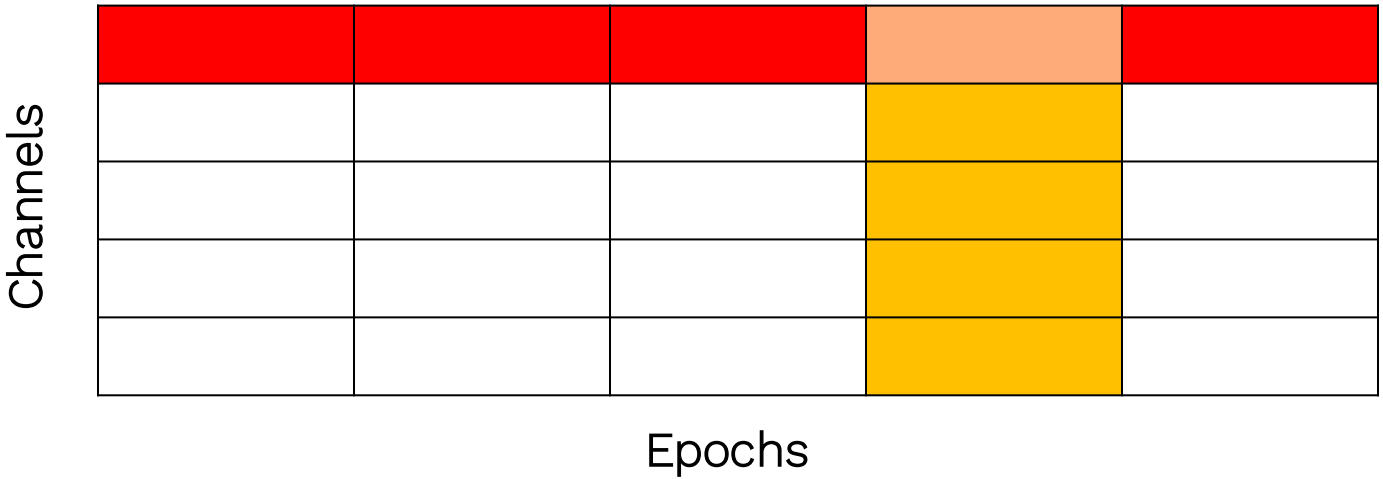
Extraphysiologic -> Remove channel or epochs

If the artifact occurs multiple times in the same channel: remove the channel from the recording

If the artifact occurs once or it involves multiple channels: remove the epoch

What to do with artifacts?

Important: You cannot remove one channel only from one epoch

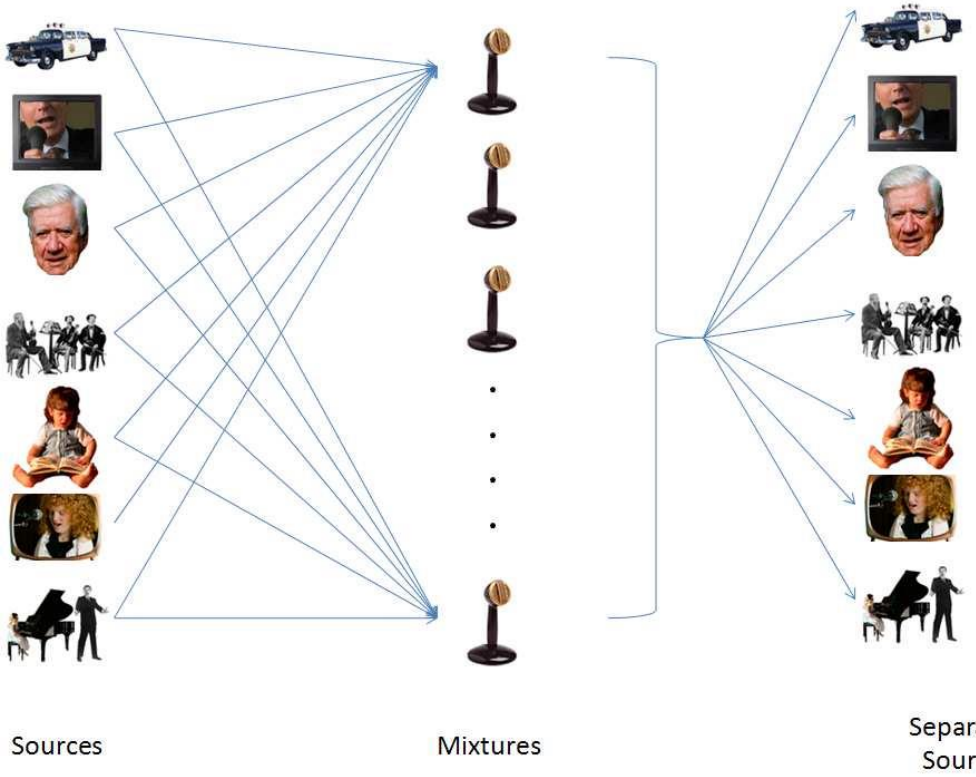


ICA

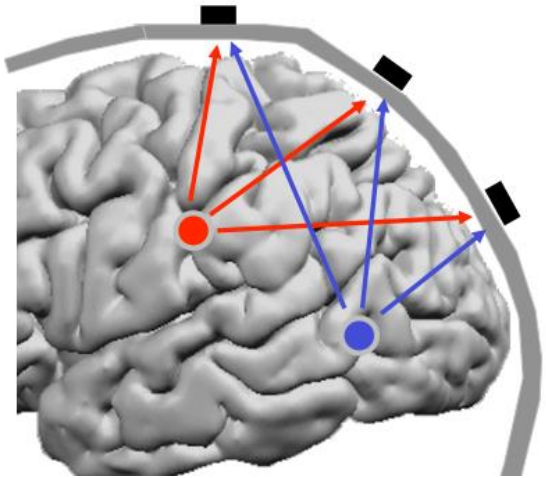
Cocktail party problem:

ICA separates the sounds

Considering the microphones position but without knowing anything about the different sound sources



ICA on EEG



EEG signals are a mixture of neural and artefactual processes -> ICA separates independent sources linearly mixed in several sensors. Each source is a **component** of the overall activity.

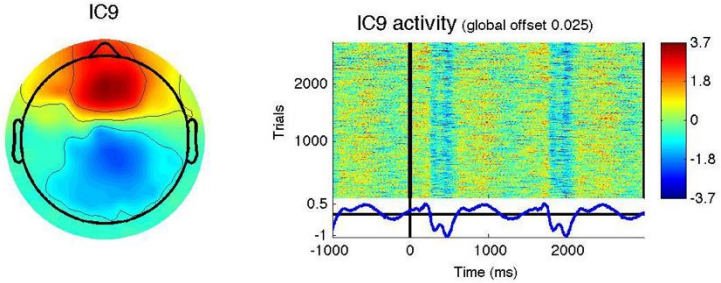
Artifactual Components (bad) are removed from the recording

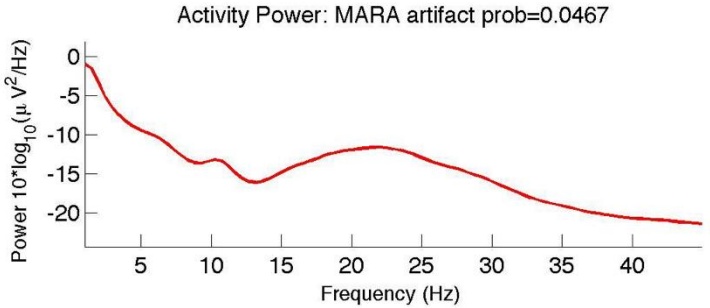
ICA Components

Neural component

(good component)

- Activity is consistent across trials
- Spectra is physiologically decreasing



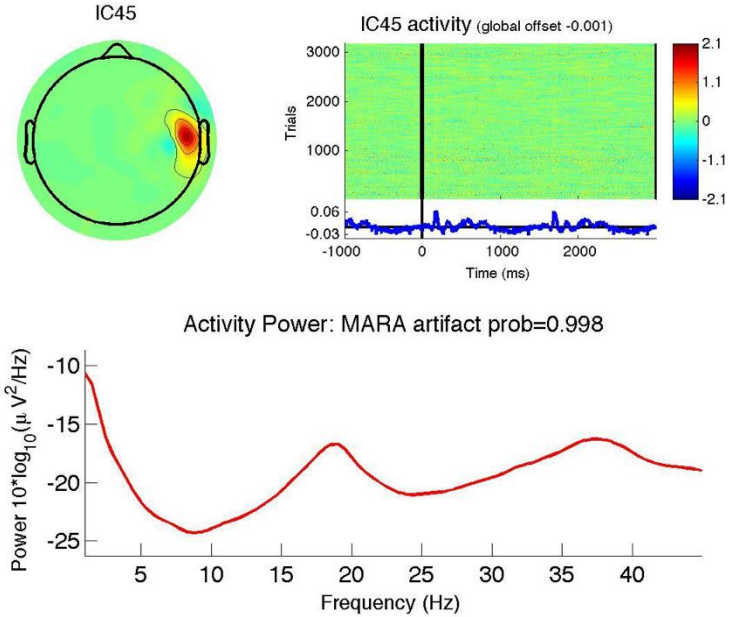


ICA Components

Muscle component

(bad component)

- Dipole in the topography
- Spectra increase in the high frequencies
- Activity is spotted and non time-locked

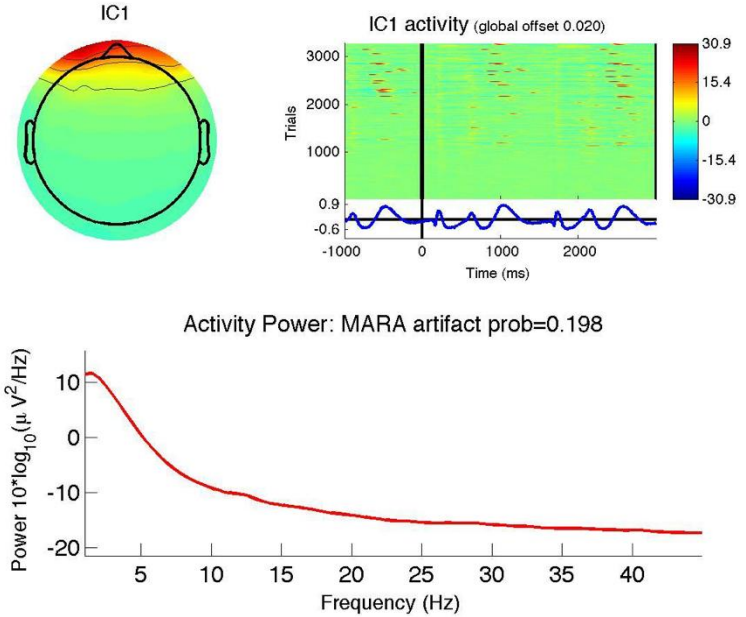


ICA Components

Eye component

(bad component)

- Activity localized near the eyes
- Bands in the time/trials plot



Channels Interpolations

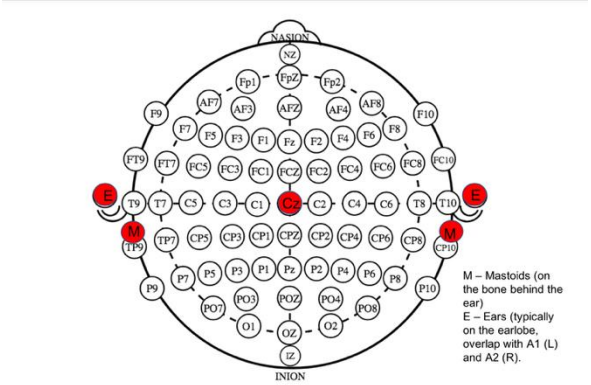
Data from missing electrodes (because of artifacts) are estimated based on the activity and locations of other electrodes.

Note: Interpolated channels do not provide unique data but a weighted sum of the activity of their neighbors

Average Reference

Possible References (theoretically anywhere):

- Mastoids
- Earlobes
- Average of all the channels



Additional materials

- Purves et al. Neuroscience chapters 1,2,5
- Introduction on EEGLAB https://www.youtube.com/watch?v=R_nf1HRScx4&list=PLXc9qfVbMMN2NksmDeqizCI1z5DJBlqC6
- Cohen, M. X. (2014). Analyzing neural time series data: theory and practice. MIT press
- Lopes da Silva, EEG and MEG: Relevance to Neuroscience. Neuron 2013
- How to label: https://labeling.ucsd.edu/tutorial/labels